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Investigation of teachers' instructional material development literacy in terms of knowledge and perception

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Abstract
This study examines teachers' material development literacy, focusing on their perceptions and levels of knowledge. Using a survey model, data were collected from 123 teachers from different disciplines and educational levels. Two data collection tools were employed to assess perceptions and theoretical knowledge, respectively (the Scale of Material Development Literacy Perception for Teachers and the Instructional Material Development Knowledge Test). The results indicate that while teachers have high levels of perception regarding material development, their knowledge levels remain medium and unevenly distributed. No significant correlation was found between these dimensions, suggesting that perceptions may be influenced by factors such as motivation or self-efficacy rather than actual knowledge. Professional development programs, although widely implemented, were found to lack the practical focus necessary to bridge this gap. Teachers with specialized training demonstrated better alignment between knowledge and practice, highlighting the importance of targeted, hands-on professional development. This study underscores the need to integrate theory and practice in teacher education programs. These results provide avenues for refining teacher training and improving classroom outcomes.

Keywords Material development literacy, Teacher education, Instructional materials

1 Introduction

Conventional pedagogical approaches, which are predominantly based on passive learning models and a reliance on rote memorization, are increasingly viewed as insufficient for addressing the multifaceted demands of contemporary education [1, 2]. Current educational paradigms emphasize the need for instructional materials that are not only content-rich but also engaging, flexible, and capable of accommodating diverse learning preferences [3–7]. Instructional materials have traditionally been sourced from repositories that were readily accessible to teachers and students. However, the demands of the modern era require their continuous revision and adaptation, which in turn requires teachers to assume greater responsibility for their design and adaptation. This transition

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highlights the need for teachers to have advanced pedagogical and technological competencies [8–10].

While materials development in teaching is defined as the process of designing, creating, and adapting instructional resources to facilitate effective teaching and learning, it entails a diverse spectrum of activities [11–13]. These activities include the creation of original teaching tools, the customization of existing resources, and the evaluation of materials to ensure congruence with pedagogical objectives and learner requirements [14, 15]. For instance, effective teaching materials should be designed in alignment with fundamental principles such as inclusiveness, cultural sensitivity, and technological integration [16, 17]. These materials should be characterized by an iterative framework encompassing phases such as analysis, design, development, implementation, and evaluation-revision [18, 19]. This approach facilitates the connection between theoretical understanding and practical implementation, promoting active engagement by fostering critical thinking skills among students, thereby achieving their intended objectives [20–22].

The development of instructional materials is recognized for its potential to improve pedagogical practices by equipping educators with tools to deliver lessons that are more effective, structured, and engaging [23–26]. The integration of appropriate materials has been demonstrated to make teaching more dynamic, adaptable, and aligned with students' needs, thereby fostering a more effective and meaningful learning process [24, 25]. However, as the teaching environment evolves, material development emerged as a pivotal aspect of teaching practice, expanding teachers' roles beyond the traditional task of delivering content to include the creation and adaptation of instructional materials [27–30]. One of the primary reasons for this is that teachers possess a comprehensive understanding of their students' unique needs and learning styles, making them the most competent to determine the characteristics of the required materials [31, 32]. This dual role positions them as both creators and users of instructional resources [33, 34]. This process not only enhances teaching practices and student learning outcomes, but it also significantly contributes to the professional development of teachers. For instance, several studies in the literature assert that teachers involved in the materials development process are better prepared and more adept at aligning resources with curriculum standards and learning objectives [24, 26, 35–37]. They are also more effective in enhancing the contextual relevance of materials by incorporating student feedback and integrating innovative, interactive elements that promote engagement. To achieve these conditions, it is essential for teachers to possess material development competencies, thereby fostering instructional material development literacy.

Material development literacy for teachers is defined as the extent of teachers' capability to effectively design, adapt, evaluate, and utilize instructional materials [38, 39]. This concept holds significant importance in improving the adaptability and efficacy of teaching processes in addressing student needs. Furthermore, it represents a critical dimension of teachers' professional expertise, enabling the alignment of learning materials with targeted educational outcomes through the integration of pedagogical, technological, and content knowledge [40–44]. These competences include key aspects such as planning and design, pedagogical expertise, adaptation and organization, technological competence, a culturally responsive and inclusive perspective, innovation and flexibility, as well as skills in evaluation and re-development [30, 38, 45–47].

In the field of education, the processes of planning and design encompass educators' capacity to develop effective and appropriate materials while addressing the multifaceted learning needs and pedagogical goals of students. In this context, pedagogical expertise is defined as educators' proficiency in the creation of materials that are both aligned with curriculum standards and conducive to enhancing students' learning processes [13, 46, 48]. The processes of adaptation and organization involve tailoring existing materials to ensure alignment with diverse grade levels, cultural frameworks, and the unique requirements of students. Essentially, adaptation and organization reflect educators' ability to innovatively adapt existing resources to address contextual demands. Technological competence pertains to teachers' proficiency in designing and implementing digital materials through the effective use of technology. A culturally responsive and inclusive perspective is defined as the creation of materials by taking into account the cultural and linguistic differences of students, thus ensuring the inclusion of all students in the learning process [49, 50]. Innovation and flexibility in education encompass educators' capacity to modify teaching materials to suit diverse contexts and to devise creative solutions [51–53]. Evaluation and re-development, on the other hand, refer to teachers' ability to analyses the effectiveness of the teaching materials they create, to implement revisions, and to foster ongoing enhancements in this process [54–56].

When the literature is examined, it becomes evident that teachers' material development literacy levels vary significantly across the globe, influenced by factors such as the socio-economic contexts of education systems, the scope of teacher education programs, and access to technology. For example, in European countries, teachers demonstrate higher proficiency in material development, attributed to greater access to digital technologies and more frequent participation in professional development programs [57–59]. In contrast, in low-income countries, the rate of material development is lower, primarily due to limited access to resources and infrastructure [30, 60, 61]. In another study conducted in Indonesia, it was revealed that teachers predominantly focused on developing materials grounded in a local context and cultural values during the material development process, while the integration of technology remained constrained [62]. In low-resource regions, such as South Africa, teachers struggle to adapt existing resources due to limited access to materials, increasing the need for creative solutions [63]. Another study in Asia reveals that in some countries, teachers have access to technology to produce advanced digital materials, but these skills are mostly concentrated in large cities [64]. Material development literacy levels are influenced by both teachers' individual skills and overarching educational policies, as well as the presence of supportive institutional structures. A study [65] highlights that professional development programs are instrumental in enhancing teachers' capacity to create innovative and interactive instructional resources. However, a study [66] state that teachers in low-income areas innovatively utilize limited resources to devise solutions, yet the challenges they encounter in this process pose significant barriers to establishing a sustainable culture of material development. Similarly, another study [59], while examining the success of teachers in technological integration processes in Europe, states that these processes are insufficient in low-income countries due to lack of resources.

Globally, the foundation for enhancing teachers' material development competencies is typically provided during teacher education processes or through in-service training programs [47, 67]. As a study [68] highlights, training programs integrated within

professional development frameworks are imperative for enhancing teachers' material development literacy. In Brazil, for instance, teachers are trained in designing digital presentations, interactive quizzes, and virtual simulations to enhance the interactivity and engagement of the learning process by integrating multimedia elements into teaching materials [69]. Similarly, in Kenya, teachers participate in workshops to develop learning materials that are appropriate to the local context, using community resources to overcome the limitations caused by limited access to ready-made materials. This approach encourages teacher innovation and enables the development of localized materials [70, 71]. In Bangladesh, teacher training programs designed to enhance EFL teaching emphasize the incorporation of pedagogical tools into the teaching process. These programs are aimed at equipping teachers with the skills to develop materials that enhance students' linguistic and communicative competencies [72]. In Indonesia, technology-based teacher education programs aim to strengthen material development skills through the use of digital tools. In particular, the programs for pre-service teachers provide them with the skills to design interactive English language teaching materials. These materials are reported to have a positive impact on student learning outcomes [73].

Notwithstanding the aforementioned examples, digital literacy plays a significant role in teacher education processes in countries with advanced technological infrastructure. For instance, in South Korea, teachers are equipped with the skills to integrate artificial intelligence and gamification into lesson planning, thereby supporting adaptive and personalized learning processes [74, 75]. In Europe, blended learning approaches are becoming increasingly important in teacher education programs. The BaCuLit Project supports secondary school teachers in improving their literacy skills and incorporate digital tools into their professional development processes through modules that combine online and face-to-face training [59]. In China, sustainable development goals are embedded within teacher education processes as part of the National Teacher Education Program. The program merges theoretical frameworks with curriculum and material development strategies in alignment with sustainable education goals [76]. These examples illustrate that different approaches can be used to increase teachers' literacy in materials development, and emphasize the importance of considering the local context and technological infrastructure in this process.

In summary, it is clear that in order to provide effective instruction, teachers must not only be users of materials but also creators. In other words, the ability to design, adapt, and evaluate instructional materials has become an essential skill for modern educators. However, there is a paucity of studies in the existing literature that examine teachers' material development literacy levels in detail [77–79]. In most of these studies, it is evident that research on teachers' material development literacy generally centers on individual perceptions (teachers' expressions of their own situations) and self-evaluations. For instance, previous studies noted the success of teachers in integrating feedback into the materials development process and creating innovative materials, although they were generally limited to teachers' perceptions, with less focus on the knowledge, and skills dimensions [80–82]. However, literacy should be approached by assessing all dimensions, including attitudes, knowledge and skills; otherwise, inaccurate results may be obtained [83–85]. In addition, it is important to emphasize that a thorough review of how educational policies and teacher development programs affect the material development processes is insufficiently represented in the literature [24, 86, 87].

Building on the previous discussion, this study aimed to examine teachers' material development literacy by exploring the perception and knowledge dimensions, which are critical components of material development literacy. In line with this purpose, the following research questions were addressed:

1) What are teachers' perception levels regarding instructional material development?
2) What are teachers' knowledge levels regarding instructional material development?
3) What is the relationship between teachers' perceptions and their knowledge levels of instructional material development?
4) What factors can enhance teachers' instructional material development literacy levels?

2 Methods

This study was designed based on the descriptive survey model to investigate teachers' instructional material development literacy levels with regard to perception and knowledge. In the study, pilot applications of the Scale of Material Development Literacy Perception for Teachers [88] and the Instructional Material Development Knowledge Test were conducted, and validity and reliability analyses were performed. Throughout the research process, the scales were administered to the participants, and general information forms were completed. The data collected were analyzed, and a comprehensive evaluation was conducted.

2.1 Participants

This study employed a convenience sampling method. An invitation was distributed through school administrative channels resulted in the voluntary participation of 123 teachers across different educational levels and subject areas, allowing access to a demographically diverse but non-random sample. The sample comprised 73 female and 50 male teachers, aged between 24 and 60 years. In terms of institutional affiliation, eight participants were employed in primary schools, 39 in secondary schools, 55 in high schools, and 21 in science and art centers. The disciplinary distribution of participants was as follows: science ($n = 15$), foreign languages ($n = 14$), mathematics ($n = 13$), information technologies ($n = 11$), religious culture ($n = 11$), social sciences ($n = 9$), primary education ($n = 8$), physical education ($n = 6$), language and literature ($n = 10$), vocational courses ($n = 5$), philosophy ($n = 4$), visual arts ($n = 4$), music ($n = 3$), special education ($n = 1$), and technology and design ($n = 9$). Participation was voluntary, and no specific inclusion or exclusion criteria beyond active teaching status were applied. Although demographic diversity was achieved, the lack of randomization limits the representativeness of the findings. Invitations were sent to approximately 180 teachers, resulting in a response rate of around 68.00%. Potential bias due to self-selection should be considered when interpreting the results.

2.2 Data collection tools

In this study, Scale of Material Development Literacy Perception for Teachers (SMDLPT) [88] was used to assess the perception component of teachers' material development literacy, and the Instructional Material Development Knowledge Test (IMDKT) was administered to determine the level of teachers' material development knowledge.

2.2.1 Scale of material development literacy perception for teachers

For the purpose of this study, the SMDLPT [88] was employed as a data collection tool. This scale was selected because it aligns with the current study's aim of evaluating teachers' perceptions across multiple facets of instructional material development. Furthermore, previous research [88] has confirmed its validity and reliability within similar educational contexts, highlighting its suitability for assessing perception-based competencies in teacher education.

The scale is designed with a four-factor structure to measure the perception levels of teaching material development competencies. The SMDLPT comprises four dimensions that reflect distinct aspects of teachers' perceptions regarding instructional material development. The Self-Adequacy factor captures teachers' perceived competence in designing and adapting instructional materials; for example, one item states, "I consider myself competent in developing teaching materials." The Desire dimension addresses teachers' motivation and interest in material development activities, as exemplified by the item, "I wish I had received better training on the instructional material development process." The Material Use factor assesses the frequency and purposefulness of using instructional materials in classroom settings, such as "I make the lesson more fun with the help of teaching materials." Lastly, the Process Skill factor focuses on teachers' awareness and application of the phases of material development—including planning, implementation, and evaluation—illustrated by the item, "I take students' readiness levels into consideration in the process of developing teaching materials." These operational definitions and sample items help clarify the conceptual structure of the scale and enhance its reproducibility for future studies.

Content and construct validity evidence were established in the original validation study. In that study, expert reviews were conducted during the item development phase to ensure content alignment with theoretical definitions of material development literacy. Exploratory and confirmatory factor analyses supported a four-factor structure (Self-Adequacy, Desire, Material Use, and Process Skill), and the total variance explained was above 60.00%. The instrument demonstrated acceptable internal consistency across dimensions supporting its structural validity. Given this prior validation, the instrument was deemed appropriate for reuse in this study without further modifications. The overall Cronbach's Alpha coefficient of the scale was calculated to be 0.91, and the Cronbach's Alpha coefficients for the individual factors were calculated to be 0.83, 0.66, 0.81, and 0.96, respectively.

The SMDLPT is a five-point Likert-type scale consisting of a total of 23 items, with items rated on a scale from strongly disagree to strongly agree. Scores obtained from the scale range from 23.00 to 115.00, and sub-factors and overall scores are presented in Table 1.

Table 1 Lower and upper score values regarding the SMDLPT and subscale levels

	Very Low	Low	Medium	High	Very High
F1	6.00–8.99	9.00–13.99	14.00–23.99	24.00–28.99	29.00–30.00
F2	4.00–5.99	6.00–8.99	9.00–15.99	16.00–18.99	19.00–20.00
F3	3.00–4.99	5.00–7.99	8.00–11.99	12.00–13.99	14.00–15.00
F4	10.00–14.99	15.00–22.99	23.00–38.99	39.00–46.99	47.00–50.00
SMDLPT	23.00–32.00	33.00–50.99	51.00–87.99	88.00–106.99	107.00–115.00

F1: Self-Adequacy, **F2**: Desire, **F3**: Material Use, **F4**: Process Skill

2.2.2 Instructional material development knowledge test

The second data collection tool was the Instructional Material Development Knowledge Test (IMDKT) developed by the researchers. This test was developed to assess teachers' level of knowledge about instructional material development processes. The test consists of 20 multiple-choice items, each with five choices, one correct and four incorrect, that assess basic concepts and theoretical models related to the instructional materials development process. The validity and reliability analyses of the IMDKT were conducted rigorously. Content validity was ensured by aligning item outcomes with the test objectives and by consulting with field experts. The complete test time ranges from 20 to 30 min, and it was administered to a sample of 123 teachers from various school types and disciplines. Construct validity was verified through item and test analyses conducted throughout the test development process.

In the reliability analysis, the Kuder-Richardson 20 (KR-20) coefficient was calculated and found to be 0.71. This value indicates that the test has an acceptable level of reliability. In addition, item difficulty and discrimination indices were evaluated, revealing that most items had high discrimination and appropriate levels of difficulty. The mean difficulty level of the test was calculated to be 0.64, indicating that the test was of moderate difficulty—neither too easy nor too difficult—and exhibited a balanced level of challenge. To categorize the scores obtained from the knowledge test, which range from 0 to 100 points, they were grouped as very low (score < 45.00), low (45.00 ≤ score < 55.00), medium (55.00 ≤ score < 70.00), high (70.00 ≤ score < 85.00), and very high (85.00 ≤ score).

Although the IMDKT was developed by the researchers, it was grounded in an established theoretical framework, and content validity was indirectly ensured by aligning the items with core principles and processes of instructional material development. The items designed to reflect widely accepted pedagogical concepts and models, and internally reviewed for clarity and representativeness. To support construct validity, item discrimination and difficulty analyses were conducted, and the test demonstrated acceptable internal consistency (KR-20 = 0.71). Given that the instrument aims to assess a distinct theoretical construct (i.e., knowledge), its weak correlation with the perception scale ($\rho = 0.03$, $p = 0.79$) is theoretically meaningful, supporting its discriminant validity.

2.3 Data collection

In this study, a quantitative data analysis approach was employed to assess teachers' literacy levels in instructional material development. The data were collected by the researcher through face-to-face, and ethical rules and scientific methods were meticulously followed. An informed consent form was presented to the participants prior to the application of the data collection tools and the participants were allowed to participate in the study on a voluntary basis.

The validity and reliability analyses of the scales and the test had been conducted prior to this study, and pilot applications were carried out in this study to verify the accuracy and applicability of the instruments. Consistent conditions were maintained for all participants during the main process, and appropriate instructions were provided to ensure correct completion of the scales.

After providing the explanation to the participants, the first set of data was collected by distributing the personal information form, which included demographic

characteristics, along with the SMDLPT. Then, the IMDKT was applied to the participants and raw data on the level of knowledge were collected.

2.4 Data analysis

The data collected in the study were analyzed using statistical methods. During the analysis, frequency and percentage distributions of demographic variables were calculated, and arithmetic means and standard deviations for perception and knowledge levels were determined. To assess teachers' knowledge regarding material development, the relationships between the scale and test scores were examined using correlation analysis. As a result, both descriptive and inferential statistics were employed in the data analysis to offer a comprehensive evaluation of teachers' perceptions and knowledge levels regarding instructional material development processes.

3 Results

3.1 Results related to teachers' perception levels of instructional material development

In order to answer the first research question of the study, "What are teachers' perception levels regarding instructional material development?", the participants' scores on the SMDLPT were analyzed, and the descriptive statistics, organized by factors, are presented in Table 2.

Upon examining the results in Table 2, it is evident that the participants' perceptions of instructional material development are high ($\bar{x}$= 95.01, sd = 15.35). When the results obtained were analyzed in the context of sub-factors, it was determined that the perceptions of the teachers in terms of the self-adequacy factor were at a medium level ($\bar{x}$= 22.06; sd = 4.89), while the perception levels of the desire ($\bar{x}$= 17.95, sd = 2.38), material use ($\bar{x}$= 13.26, sd = 1.63) and process skill ($\bar{x}$= 41.73, sd = 9.64) were high. An analysis of the participants' scores on the scale revealed that 56.10% of the participants reported high perception levels (n = 69), 22.00% reported medium perception levels (n = 27), 19.50% reported very high perception levels (n = 24), and 2.40% reported low perception levels (n = 3).

3.2 Results related to the knowledge dimension of teachers' level of instructional material development

In line with the second research question of the study, teachers' knowledge regarding instructional material development was assessed using the IMDKT. As a result of the analyses, it was found that most of the participants (n = 37) demonstrated a medium level of knowledge ($\bar{x}$= 60.41, sd = 3.97), while 27.60% demonstrated high knowledge (n = 34, $\bar{x}$= 74.26, sd = 4.11), 17.10% demonstrated very low knowledge (n = 21, $\bar{x}$= 31.43, sd = 9.37), 13.80% demonstrated low knowledge (n = 17, $\bar{x}$= 47.94, sd = 5.54), and 11.40% demonstrated very high knowledge (n = 14, $\bar{x}$= 87.14, sd = 3.23). In addition, when the

Table 2 Descriptive statistical results regarding the SMDLPT levels of the participants in the study

Sub-factor	Level	Min level	Max level	– $\bar{x}$	Sd
Self-Adequacy	Medium	7.00	30.00	22.06	4.89
Desire	High	10.00	20.00	17.95	2.38
Material use	High	9.00	15.00	13.26	1.63
Process skill	High	10.00	50.00	41.73	9.64
Total SMDLPT	High	44.00	115.00	95.01	15.35

knowledge test results of all participants were analyzed, it was found that the mean knowledge level of the participants was at a medium level ($\bar{x}$=60.61, sd=18.09).

3.3 Results on the relationship between teachers' perception levels of instructional material development and their knowledge levels

To examine the relationship between teachers' perceptions and knowledge levels regarding instructional material development, the correlation between the scores of the SMDLPT and the IMDKT was calculated, and no significant relationship was found ($\rho = 0.03$; $p = 0.79$). For a more in-depth analysis, the participants' scores on the SMDLPT and the IMDKT were re-analyzed by categorizing them based on their perception levels. It was found that there was no significant relationship between the low ($n = 3$; $\rho = 0.50$; $p = 0.67$), medium ($n = 27$; $\rho = 0.20$; $p = 0.31$), high ($n = 69$; $\rho = 0.01$; $p = 0.95$), and very high ($n = 24$; $\rho = -0.26$; $p = 0.21$) perception levels and the IMDKT.

3.4 Results related to teachers' instructional material development literacy levels

In order to clearly determine teachers' literacy level, participants' perception and test scores were grouped by level. The perception and achievement levels were then analyzed using cross-analysis, and the results are presented in Tables 3 and 4, including both lower and upper categories in the comparison.

Analyzing Table 3, it can be concluded that 27.64% ($n = 34$) of the participants were able to correctly identify the perception level according to the performance level. When the calculation is done again, taking into account the total margin of error, it is found that only 65.85% ($n = 81$) of the participants could correctly define their levels.

When the literacy levels of the teachers ($n = 81$), whose levels were correctly identified when the margin of error was considered, were analyzed in terms of perception and knowledge, it was found that 16.05% ($n = 3$) were at very high level, 41.98% ($n = 34$) were at high level, 34.57% ($n = 28$) were at medium level, 6.17% ($n = 5$) were at low level and 1.23% ($n = 1$) were at very low level instructional material development literate.

In order to further investigate the unclear results and to address this issue in accordance with the main objective of the research, the comparison and cross-tabulation of the scores from the perception scale and the achievement test are presented in more detail in Table 4. According to the results of the analysis, it is seen that the scores of 123

Table 3 Results related to the cross-examination of the participants' and IMDKT levels

		IMDKT Levels.					
		Very Low	Low	Medium	High	Very High	Total (n)
SMDLPT Levels	Very Low	0*	0	0	0	0	0
	Low	1	1*	1	0	0	3
	Medium	5	4	8*	9	1	27
	High	11	7	19	22*	10	69
	Very High	4	5	9	3	3*	24
	Total	21	17	37	34	14	123
One-to-one Match total*		0	1	8	22	3	34
Includes the total margin of error		1	5	28	34	13	81

Model 1: Unadjusted;

Table 4 Results related to the cross examination of participants' SMDLPT scores and IMDKT scores

SMDLPT	IMDKT 5	10	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	Total
44	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1
47	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
49	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
52	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	2
59	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
61	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
64	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1
65	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
66	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1
67	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1
70	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
71	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
81	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
82	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1
83	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
84	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	2
85	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1
86	0	0	0	0	0	0	1	0	0	0	0	3	1	0	1	0	0	6
87	0	0	0	0	0	0	1	0	0	2	1	1	0	0	0	0	0	5
88	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	3
90	0	0	0	0	1	0	0	0	1	1	2	0	0	0	0	0	1	6
91	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
92	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	2
93	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1
94	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	1	0	5
95	0	0	0	0	0	0	0	1	0	1	0	0	1	0	0	0	0	3
96	0	0	0	0	0	0	0	0	0	1	0	0	3	1	0	0	0	5
97	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
98	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1
99	0	0	0	0	1	1	0	0	1	0	0	2	0	0	1	0	0	6
100	0	0	0	0	0	1	0	1	0	1	0	1	0	0	1	0	0	5
101	0	0	0	0	0	0	0	1	0	1	1	0	1	1	0	0	0	5
102	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	3
103	0	0	1	0	1	0	0	0	0	0	0	1	0	0	1	0	0	4
104	1	0	0	0	1	0	0	0	0	0	0	0	2	2	0	1	0	7
105	0	0	0	1	0	1	0	1	0	1	2	0	0	0	1	1	0	8
106	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	3
107	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	1	0	5
108	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	2
109	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	2
110	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	2
111	0	0	0	2	0	0	0	0	0	0	1	0	0	0	0	0	0	3
112	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	2
113	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
114	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
115	0	0	0	0	0	0	2	0	2	1	0	0	0	0	1	0	0	6
Total	1	1	2	6	5	6	7	10	10	14	13	14	11	9	9	4	1	123

ROC, Receiver Operating Characteristic; ACAG, albumin-corrected anion gap; AG, anion gap

participants ranged between 44.00 and 115.00 on the SMDLPT, and from 5.00 to 95.00 on the IMDKT. When the margin of error was widened (one lower-one upper value was included), 4.92% ($n = 6$) were at very high level, 30.33% ($n = 37$) at high level,14.75% ($n = 18$) at medium level, 1.64% ($n = 2$) were found to be literate in the dimension of perception and knowledge of instructional material development at a low level; only 51.64% ($n = 65$) of the total participants ($n = 123$) were able to correctly align their perception and knowledge levels of perception according to the level of knowledge.

The prevailing notion is that an increase in SMDLPT scores is concomitant with an increase in IMDKT scores; however, this relationship was not found to be linear in all cases. For example, two of the eight participants with an SMDLPT score of 105.00

scored 85.00 or higher on the IMDKT. In contrast, the IMDKT scores of the six participants with a SMDLPT score of 99.00 exhibited a wide distribution ranging from 35.00 to 80.00.

4 Discussion

In this study, the perception and knowledge dimensions of teachers' material development literacy were investigated. Previous studies [82, 89–91] on the level of teachers' instructional material development emphasize the effect of material development on teachers' professional competencies by examining it generally in the perception dimension or partially in the knowledge dimension. This study, however, found no significant relationship between perception and knowledge levels, suggesting that these two dimensions should be evaluated independently.

The results indicate that teachers generally had a high level of perception regarding the development of instructional materials, but had a medium level of perception in the self-adequacy subfactor. Although this situation can be interpreted as a lack of self-efficacy in teachers' perceptions of developing instructional materials, the results revealed that they had a very high level of literacy perception. Although some studies in the literature [24, 92–94] suggest that perceptions of self-adequacy are directly related to teachers' current knowledge and skills in the use of technology and materials, no significant results were obtained in this study. Teachers scored lower in the Self-Adequacy subfactor compared to other factors, which was attributed to their reliance on ready-made materials in their professional practice, resulting in limited material development and insufficient emphasis on these skills in pre-service teacher training [95, 96].

The desire dimension was perceived at a high level. This shows that teachers are highly motivated in the process of material development. Some studies emphasize that teachers' motivation is a critical factor in the adoption of pedagogical innovations and material development processes [24]. In particular, increasing motivational dispositions enable teachers to be more involved in material development processes, while at the same time increasing their professional satisfaction [93].

The high perception of the material use dimension suggests that teachers believe they can use available materials in line with their pedagogical purposes. However, studies in the literature suggest that teachers often struggle to select and use appropriate materials according to specific strategies [97–99]. Given that the ability to use materials is a fundamental element of effective classroom management and addressing individual learning needs [38, 46], it is likely that existing research on this topic does not fully capture the current situation.

The elevated perception of the Process Skill sub-dimension indicates that teachers possess a strong conviction in their analytical and creative aptitude in material development processes. Despite the prevailing literature [41] emphasizing the correlation between these skills and teachers' capacity to design innovative and student-centered materials, this perception of learners has been found to be inaccurate in actual settings, as elaborated upon in the following paragraph. Given the established role of process skills in enhancing teachers' capacity to evaluate, transform, and reconstruct materials [56], this discrepancy is regarded as a significant result.

The research indicates that teachers possess a medium level of theoretical knowledge about material development. However, the fact that the distribution is quite varied

shows that there are various factors that influence teachers' material development literacy. Although the results showed that teachers who participated in professional development programs had a higher level of knowledge about material development literacy, the fact that the majority of participants had a medium level of knowledge was interpreted to mean that the professional development programs were not effective enough. Based on participant feedback, the fact that the training they received before or after beginning their careers remained at a theoretical level rather than being practical, and that it included more general content rather than field-specific training, was seen as the main reason for this low level. In addition, while the heterogeneous knowledge levels may be linked to teachers' educational backgrounds, branches, and professional experience, the notable proportion of teachers with very low knowledge levels suggests a lack of basic knowledge of material development processes. This uneven distribution also highlights the difficulty of ensuring consistent competence in materials development. Teachers often rely on personal initiative or contextual learning to develop these skills, which can lead to inconsistent levels of skill across the profession [100, 101]. This level of inconsistency is attributed to the fact that issues such as theoretical models, pedagogical practices, and technology integration related to the material development process are not sufficiently addressed in current educational processes [102, 103]. Previous studies [104, 105] supporting this interpretation emphasized that participation in professional development programs is crucial for enhancing teachers' material development skills. However, it was stated that only theoretical trainings may be insufficient to improve teachers' practical application skills.

Although the number of teachers with high theoretical knowledge is above a certain proportion in the study, the general results suggest that these teachers may still face challenges in implementing effective classroom practices. This interpretation is consistent with the concept of pedagogical content knowledge, which emphasizes the integration of content knowledge and pedagogical strategies to create meaningful learning experiences [86, 106].

The lack of a significant correlation between participants' perception and knowledge levels suggests that these two dimensions may be influenced by various external factors, warranting further investigation. The results indicate that while teachers generally have high perception levels of instructional material development, their knowledge levels do not align with this perception. This suggests that although teachers have a positive attitude towards the process of material development, they lack conceptual knowledge and practical application skills.

These results, which indicate no relationship between perception and knowledge levels, are considered to be related to Self-efficacy Theory [107] and Self-determination Theory [108]. Self-efficacy Theory [107] emphasizes that individuals' self-efficacy perceptions can be independent of their actual knowledge and skill levels. In this context, teachers' high perception levels of material development processes may not accurately reflect their actual knowledge in this field. Similarly, Self-determination Theory [108] suggests that individuals' motivations and self-evaluations may not always accurately reflect their level of knowledge. Teachers who exhibit medium self-efficacy, for instance, may possess adequate knowledge but lack the confidence to fully leverage their skills in material development. Conversely, high self-efficacy can motivate teachers to engage more actively with material development processes, potentially leading to incremental

knowledge gains through practice. This lack of relationship can also be explained by the Dunning-Kruger Effect [109, 110]. According to this effect, individuals with low levels of knowledge tend to overestimate their skills. This may explain why teachers have high perception levels despite lacking knowledge about material development.

This weak relationship between perception and knowledge levels shows that there is a disconnect between teachers' theoretical knowledge of the material development process and their competencies in practice. A similar study [111] indicated that while teachers exhibited high self-confidence in material development processes, they lacked the necessary practical competencies. This situation can be explained by the fact that teachers have positive perceptions of the process, but do not have the necessary knowledge and skills.

In summary, while the results underline a generally positive perception of material development literacy among teachers, they point to a discrepancy between their self-perceived skill and their actual level of knowledge. These gaps between perception and knowledge suggest the need for professional development programs that combine theoretical training with practical applications. In particular, the fact that existing theoretical training was not specific to the field was seen as a serious barrier. Feedback from participants and supporting literature emphasized the need for such training to be field-specific, continuous, and regularly updated. This is important because professional development programs that integrate content knowledge with disciplinary strategies significantly enhance teachers' material development skills. For example, targeted professional development in STEM for bilingual/ESL teachers showed improved content knowledge and instructional quality [112]. Similarly, programs focused on content literacy instruction emphasized the need for scaffolding and explicit integration of skills with real-world applications [113]. Collaboration among educators fosters a supportive ecosystem for developing instructional materials. This approach emphasizes the importance of teamwork and shared learning goals in professional settings to enhance literacy competence [114].

5 Conclusion

This research highlights the critical and nuanced role of teachers' material development literacy in advancing educational practices. The results reveal that, while teachers generally perceive themselves as capable in material development, a significant gap exists between their perceived competence and actual knowledge levels. This disparity underscores the complex interplay between perception, knowledge, and practical application.

The study further illustrates that professional development programs aimed at enhancing material development literacy are often inadequately aligned with teachers' practical needs and specific instructional contexts. Teachers' theoretical knowledge was found to be unevenly distributed, with significant variations based on socio-economic and infrastructural differences, suggesting a systemic issue in the provision of resources and training. Despite the high levels of motivation and positive perceptions regarding material usage and process skills, the absence of a significant correlation between perception and knowledge levels indicates that these are influenced by independent and potentially external factors.

This disconnect has critical implications for both the design of teacher education programs and the structuring of ongoing professional development initiatives. The research

thus contributes a valuable perspective to the discourse on educational reform, advocating for holistic, context-sensitive, and practice-oriented strategies to bridge the gap between perception and competence in instructional material development.

5.1 Limitation

Despite the meticulous design of the research, certain limitations must be acknowledged. Primarily, the study was conducted in a specific region, which may limit the generalizability of its results to areas with differing educational contexts. While teachers from various disciplines participated, variations in their educational backgrounds and professional experiences might have influenced the results. The knowledge test primarily assessed theoretical understanding, offering limited insights into teachers' practical competencies in instructional material development.

Moreover, as a cross-sectional study, it did not track longitudinal changes in teachers' material development skills or assess the long-term effects of interventions. Differences in participants' access to and familiarity with digital tools may have impacted their ability to create technologically advanced instructional resources. Additionally, the study focused largely on teachers' perceptions and knowledge, potentially overlooking other critical factors, such as practical skills, attitudes, and systemic barriers. The absence of longitudinal data further limited a comprehensive evaluation of the long-term effectiveness of professional development programs in fostering sustainable improvements.

5.2 Suggestions

Given the variation in research results, it is recommended that teachers' material development literacy be assessed across multiple dimensions—perception, knowledge, and skills—rather than through a single measurement tool or dimension.

Training programs for teachers should combine theory with hands-on practice and be customized to align with specific subjects and teaching contexts. Furthermore, these programs should prioritize the integration of contemporary technologies, such as gamification and blended learning, to enhance teaching methodologies. Teachers should be provided with opportunities to work together to design, critique, and improve instructional materials. Platforms for sharing teaching resources should be established to allow for the exchange and adaptation of materials.

Teachers should receive regular feedback on their instructional materials development skills through evaluation systems. Including feedback from students can help teachers create materials that are better suited to learners' needs. Teachers should be encouraged to develop materials that reflect local cultures and contexts, thereby ensuring inclusivity and relatability for students.

Professional development programs ought to address the unique needs of specific subjects or disciplines, ensuring the training's direct applicability to teachers' classroom practices. Teachers should be encouraged to see material development as a continuous learning process, helping them improve their skills over time.

5.3 Future research

In light of the study's limitations, it is recommended that an examination be conducted of the material development literacy of teachers in various subject areas, with the aim of ascertaining the impact of special education needs. Furthermore, it is advised that the

long-term impact of professional development programs be re-examined with a focus on teachers' material development literacy, including perceptions, theoretical knowledge, and skills. Finally, it is suggested that a knowledge test should be developed to determine at which stage of material development literacy teachers have more limited knowledge, and this should also be measured using a skills test to determine the limitations.

Author contributions
This research was conducted as part of TT's thesis under the supervision of SD, with both authors contributing to the research design. TTdrafted the initial version of the article, which SD reviewed and provided feedback on. Both authors read, revised, and approved the final manuscript.

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Data availability
Data will be made available on request.

Declarations

Ethics approval and consent to participate
The study was conducted in accordance with the relevant guidelines and regulations, and it was approved by the Social Science Ethics Board of Cukurova University. Since all participants were over 18 years of age, informed written consent to participate in the study was obtained from all participants.

Consent for publication
As all participants were over 18 years of age, written informed consent for the publication of the study findings was obtained from each participant.

Competing interests
Serkan Dinçer serves on the editorial board of Discover Education and was not involved in the review or decision-making for this manuscript. Toygun Türközü declares no competing interests.

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